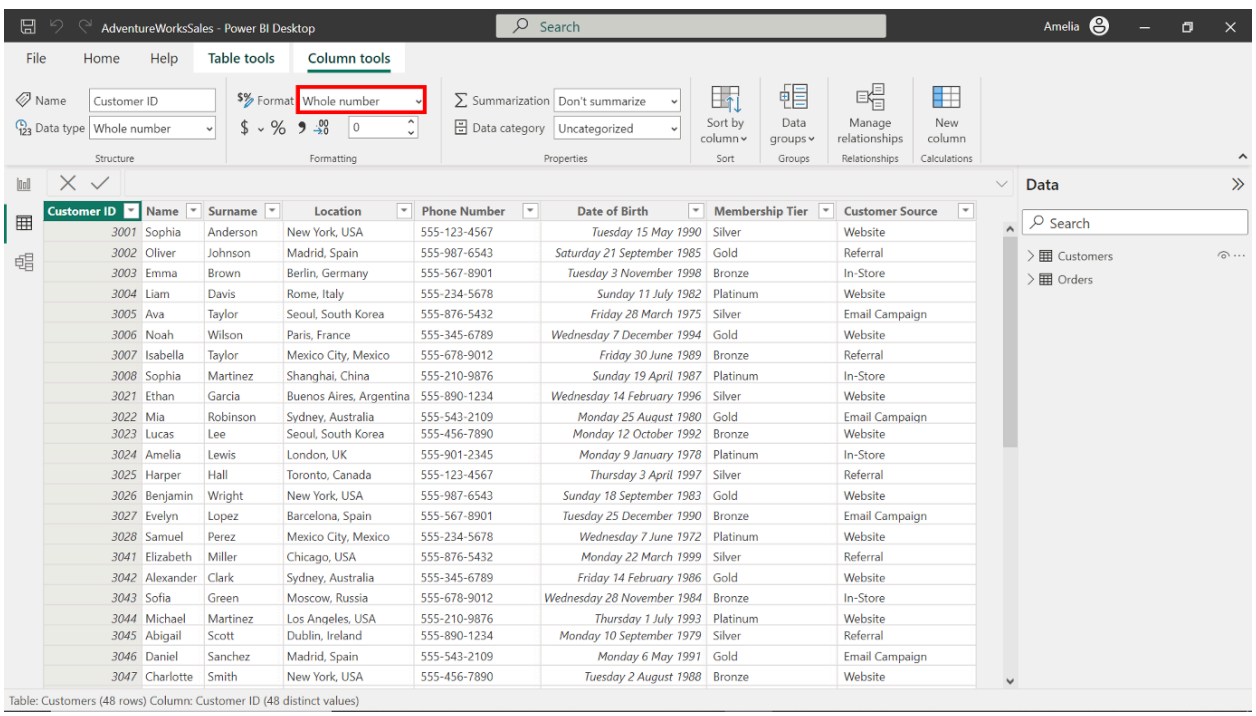
Overview

Built and optimized a data model for Adventure Works in Power BI by assigning correct data types across two tables, establishing a relationship between them, and configuring settings to improve performance and reporting control.

Step 1: Optimizing data types.

Optimizing Data Types for Customers Table

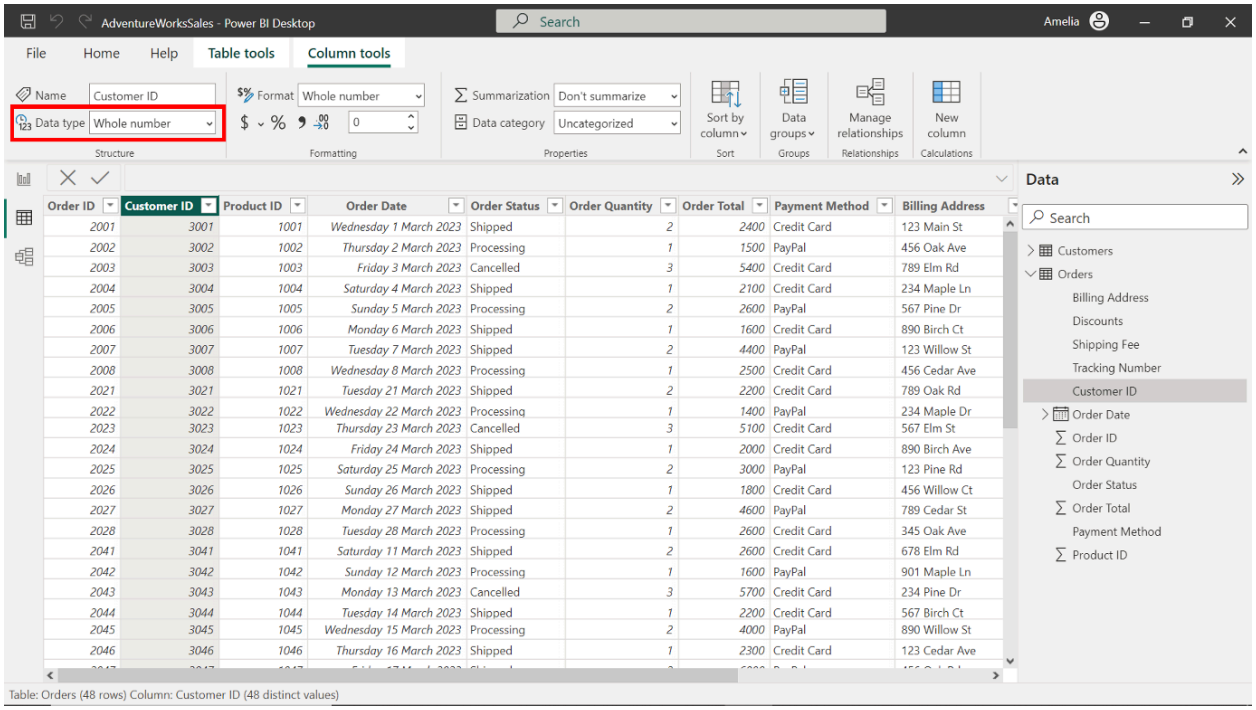
Assigned appropriate data types to all columns in the Customers table setting Customer ID as whole number, Date of Birth as date, and all remaining descriptive fields as text.
Example:



Customer ID	Name	Surname	Location	Phone Number	Date of Birth	Membership Tier	Customer Source
3001	Sophia	Anderson	New York, USA	555-123-4567	Tuesday 15 May 1990	Silver	Website
3002	Oliver	Johnson	Madrid, Spain	555-987-6543	Saturday 21 September 1985	Gold	Referral
3003	Emma	Brown	Berlin, Germany	555-567-8901	Tuesday 3 November 1998	Bronze	In-Store
3004	Liam	Davis	Rome, Italy	555-234-5678	Sunday 11 July 1982	Platinum	Website
3005	Ava	Taylor	Seoul, South Korea	555-876-5432	Friday 28 March 1975	Silver	Email Campaign
3006	Noah	Wilson	Paris, France	555-345-6789	Wednesday 7 December 1994	Gold	Website
3007	Isabella	Taylor	Mexico City, Mexico	555-678-9012	Friday 30 June 1989	Bronze	Referral
3008	Sophia	Martinez	Shanghai, China	555-210-9876	Sunday 19 April 1987	Platinum	In-Store
3021	Ethan	Garcia	Buenos Aires, Argentina	555-890-1234	Wednesday 14 February 1996	Silver	Website
3022	Mia	Robinson	Sydney, Australia	555-543-2109	Monday 25 August 1980	Gold	Email Campaign
3023	Lucas	Lee	Seoul, South Korea	555-456-7890	Monday 12 October 1992	Bronze	Website
3024	Amelia	Lewis	London, UK	555-901-2345	Monday 9 January 1978	Platinum	In-Store
3025	Harper	Hall	Toronto, Canada	555-123-4567	Thursday 3 April 1997	Silver	Referral
3026	Benjamin	Wright	New York, USA	555-987-6543	Sunday 18 September 1983	Gold	Website
3027	Evelyn	Lopez	Barcelona, Spain	555-567-8901	Tuesday 25 December 1990	Bronze	Email Campaign
3028	Samuel	Perez	Mexico City, Mexico	555-234-5678	Wednesday 7 June 1972	Platinum	Website
3041	Elizabeth	Miller	Chicago, USA	555-876-5432	Monday 22 March 1999	Silver	Referral
3042	Alexander	Clark	Sydney, Australia	555-345-6789	Friday 14 February 1986	Gold	Website
3043	Sofia	Green	Moscow, Russia	555-678-9012	Wednesday 28 November 1984	Bronze	In-Store
3044	Michael	Martinez	Los Angeles, USA	555-210-9876	Thursday 1 July 1993	Platinum	Website
3045	Abigail	Scott	Dublin, Ireland	555-890-1234	Monday 10 September 1979	Silver	Referral
3046	Daniel	Sanchez	Madrid, Spain	555-543-2109	Monday 6 May 1991	Gold	Email Campaign
3047	Charlotte	Smith	New York, USA	555-456-7890	Tuesday 2 August 1988	Bronze	Website

Optimizing Data Types for the Orders table

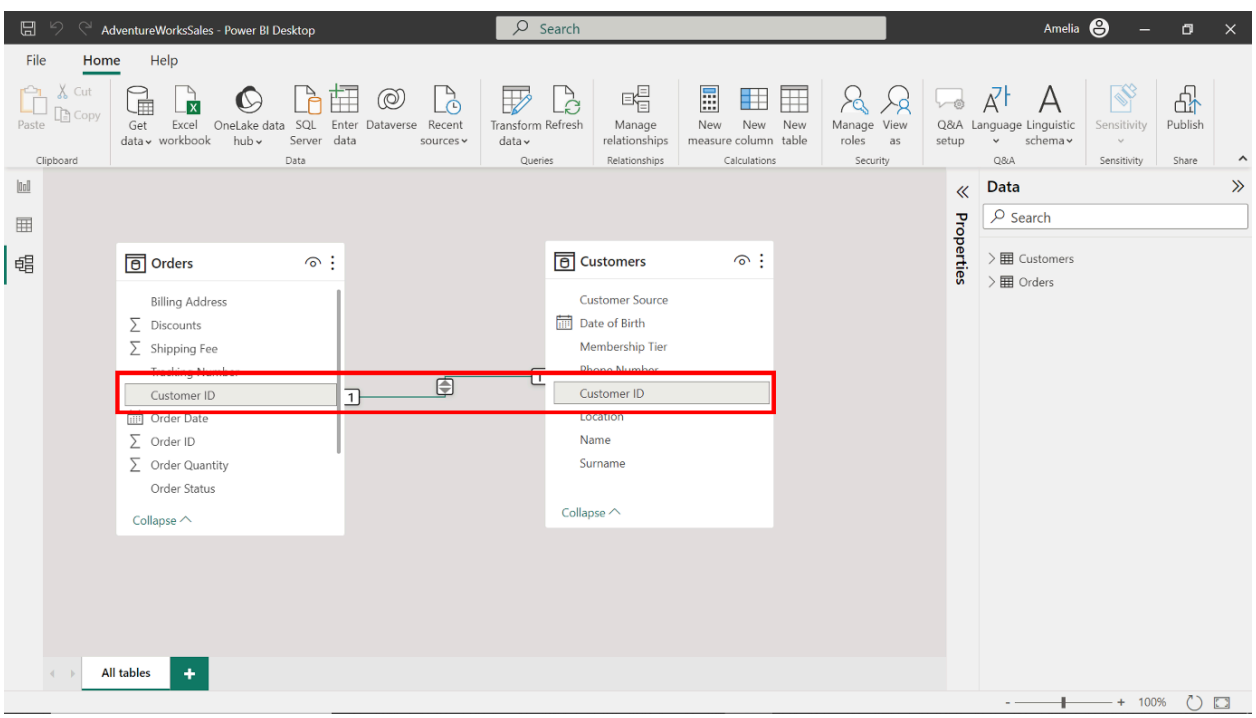
Assigned appropriate data types to all columns in the Orders table: setting Order ID, Customer ID, Product ID, and Order Quantity as whole numbers, Order Total, Discounts, and Shipping Fee as decimal numbers, Order Date as date, and all remaining fields as text.
Example:



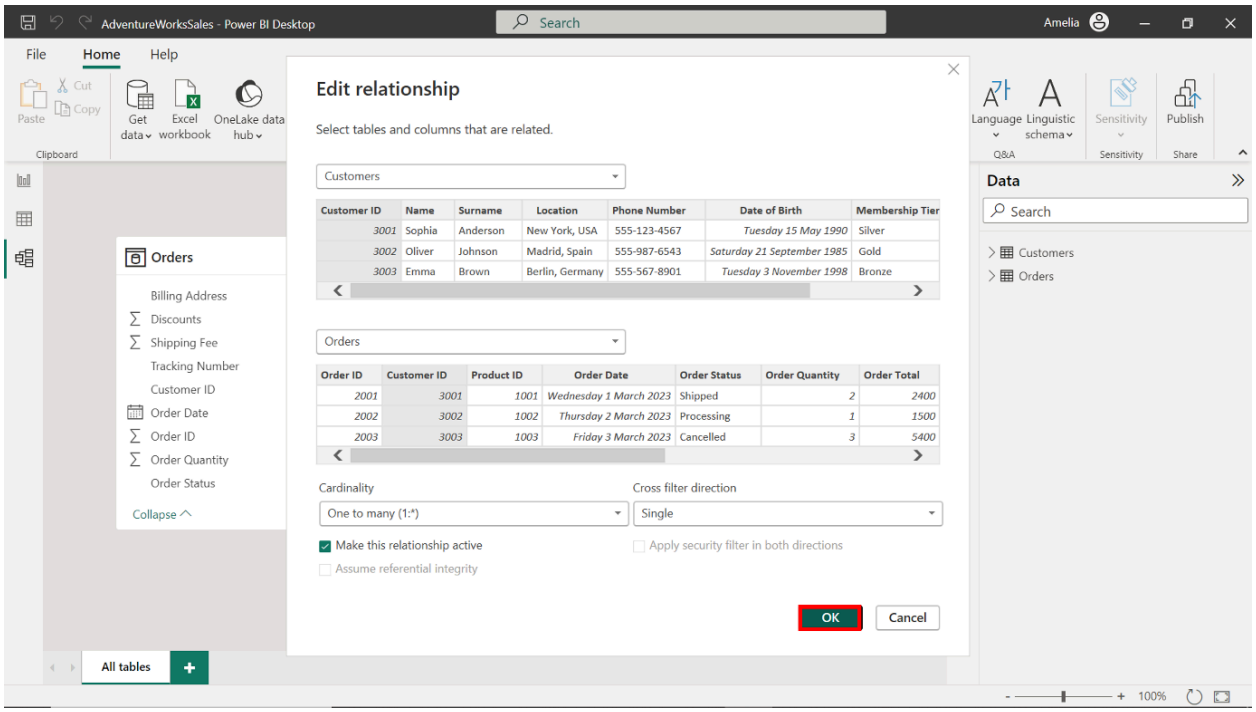
Order ID	Customer ID	Product ID	Order Date	Order Status	Order Quantity	Order Total	Payment Method	Billing Address
2001	3001	1001	Wednesday 1 March 2023	Shipped	2	2400	Credit Card	123 Main St
2002	3002	1002	Thursday 2 March 2023	Processing	1	1500	PayPal	456 Oak Ave
2003	3003	1003	Friday 3 March 2023	Cancelled	3	5400	Credit Card	789 Elm Rd
2004	3004	1004	Saturday 4 March 2023	Shipped	1	2100	Credit Card	234 Maple Ln
2005	3005	1005	Sunday 5 March 2023	Processing	2	2600	PayPal	567 Pine Dr
2006	3006	1006	Monday 6 March 2023	Shipped	1	1600	Credit Card	890 Birch Ct
2007	3007	1007	Tuesday 7 March 2023	Shipped	2	4400	PayPal	123 Willow St
2008	3008	1008	Wednesday 8 March 2023	Processing	1	2500	Credit Card	456 Cedar Ave
2021	3021	1021	Tuesday 21 March 2023	Shipped	2	2200	Credit Card	789 Oak Rd
2022	3022	1022	Wednesday 22 March 2023	Processing	1	1400	PayPal	234 Maple Dr
2023	3023	1023	Thursday 23 March 2023	Cancelled	3	5100	Credit Card	567 Elm St
2024	3024	1024	Friday 24 March 2023	Shipped	1	2000	Credit Card	890 Birch Ave
2025	3025	1025	Saturday 25 March 2023	Processing	2	3000	PayPal	123 Pine Rd
2026	3026	1026	Sunday 26 March 2023	Shipped	1	1800	Credit Card	456 Willow Ct
2027	3027	1027	Monday 27 March 2023	Shipped	2	4600	PayPal	789 Cedar St
2028	3028	1028	Tuesday 28 March 2023	Processing	1	2600	Credit Card	345 Oak Ave
2041	3041	1041	Saturday 11 March 2023	Shipped	2	2600	Credit Card	678 Elm Rd
2042	3042	1042	Sunday 12 March 2023	Processing	1	1600	PayPal	901 Maple Ln
2043	3043	1043	Monday 13 March 2023	Cancelled	3	5700	Credit Card	234 Pine Dr
2044	3044	1044	Tuesday 14 March 2023	Shipped	1	2200	Credit Card	567 Birch Ct
2045	3045	1045	Wednesday 15 March 2023	Processing	2	4000	PayPal	890 Willow St
2046	3046	1046	Thursday 16 March 2023	Shipped	1	2300	Credit Card	123 Cedar Ave

Step 2: Build and configure the model relationships.

Switched to Model View and established a relationship between the two tables by dragging the Customer ID field from the Customers table to the Customer ID field in the Orders table.

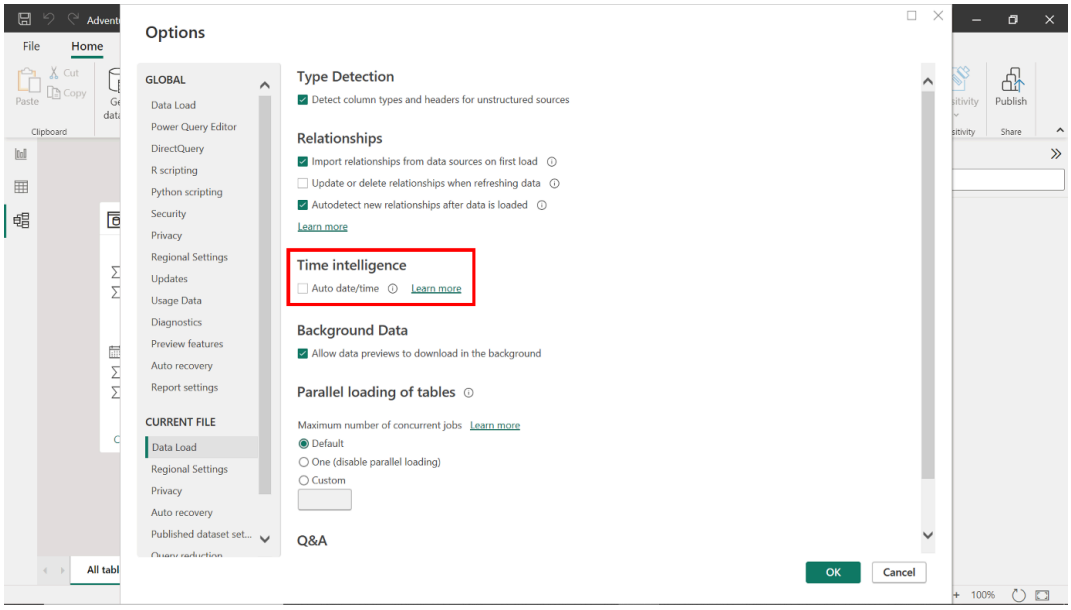


Configured the relationship as one-to-many with a single cross-filter direction, reflecting that one customer can have multiple orders and keeping the model efficient for data retrieval.



Step 3: Disable Auto/Date Time

Disabled the Auto Date/Time feature under File → Options to give greater manual control over how date fields behave in reports.



Conclusion

Produced a clean, optimized data model with correctly typed columns, a defined table relationship, and performance-focused settings, laying a solid foundation for accurate and efficient reporting in Power BI.